Linux File Permissions & `chmod`

In Linux, file permissions control **who can read, write, or execute** a file or directory.

Each file/directory has three types of permissions and three categories of users.

🔹 Types of Permissions

• **Read (r)**
• Files → allows reading the contents.

• Directories → allows listing its contents.

• **Write (w)**

• Files → allows modifying or deleting contents.

• Directories → allows creating, deleting, or renaming files inside.

• **Execute (x)**

• Files → allows running it as a program/script.
• Directories → allows entering (`cd`) into it.

🔹 Permission Groups

• **Owner (u)** → The user who owns the file.
• **Group (g)** → Users in the file's group.
• **Others (o)** → Everyone else.

Permissions are represented in a 10-character string:

```
-rwxr-xr--
```

- First char → type (`-` = file, `d` = directory). - Next 3 → owner permissions. - Next 3 → group permissions. - Last 3 → others permissions.

🔹 Example Breakdown

```
$ ls -l script.sh
-rwxr-xr--  1 prashant users  1024 Sep 25 19:00 script.sh
```

- `-` → regular file - `rwx` → owner has **read, write, execute** - `r-x` → group has **read, execute** - `r--` → others have **read only**

🔹 Numeric Representation

Permissions can also be set using numbers (octal):

Value	Permission	Meaning
7	rwx	full access
6	rw-	read + write
5	r-x	read + execute
4	r--	read only
3	-wx	write + execute
2	-w-	write only
1	--x	execute only
0	---	no access

Example: - `chmod 777 file` → everyone can do anything (⚠️unsafe). - `chmod 644 file` → owner can edit, others can read. - `chmod 755 dir` → owner full, others can list/enter.

🔹 Common Use Cases

1. Check Current Permissions

```
$ ls -l notes.txt
-rw-r--r--  1 prashant users  1024 Sep 25 19:00 notes.txt
```

- Owner → `rw-` - Group → `r--` - Others → `r--` 🌐Only the owner can edit; others can just read.

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2. Give Execute Permission

```
$ echo "echo Hello World" > hello.sh
$ ls -l hello.sh
-rw-r--r--  hello.sh

$ chmod +x hello.sh
$ ls -l hello.sh
-rwxr-xr-x  hello.sh
```

```
$ ./hello.sh
Hello World
```

♓Now the script is executable.

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3. Restrict Access (Make Private)

```
$ ls -l secrets.txt
-rw-r--r--  secrets.txt

$ chmod 600 secrets.txt
$ ls -l secrets.txt
-rw-------  secrets.txt
```

♓Only owner can read/write; no one else has access.

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4. Share Executable with Others

```
$ chmod 755 app.sh
$ ls -l app.sh
-rwxr-xr-x  app.sh
```

♓Owner has full control, others can run it.

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5. Allow Group Members to Edit

```
$ chmod 664 notes.txt
$ ls -l notes.txt
-rw-rw-r--  notes.txt
```

♓Owner and group can read/write; others can only read.

🔹 Useful `chmod` Examples

```
$ chmod +x file.sh          # Add execute permission
$ chmod -x file.sh          # Remove execute permission
$ chmod 755 script.sh       # Owner: rwx, Group: r-x, Others: r-x
$ chmod 644 document.txt    # Owner: rw-, Group: r--, Others: r--
$ chmod 600 private.txt     # Owner: rw-, Group: ---, Others: ---
$ chmod 777 public.txt      # Everyone: rwx (not safe)
```

```
$ chmod 400 read-only.txt   # Owner: r-- only
$ chmod 200 write-only.txt  # Owner: -w- only
$ chmod 111 exec-only.sh    # Everyone execute only

$ chmod u+x file.sh         # Add execute for owner
$ chmod g+w file.txt        # Add write for group
$ chmod o-r file.txt        # Remove read for others
$ chmod u=rwx,g=rx,o=r file.txt  # Set specific perms

$ chmod ugo-rwx file.txt    # Remove all permissions
$ chmod ugo+rwx file.txt    # Give all permissions
$ chmod -R 755 /path/dir    # Recursive for all inside dir
$ chmod --reference php java  # Copy perms from php → java
```

🔹 Default Permissions & `umask`

When a new file/directory is created, Linux applies **default permissions** filtered by `umask`.

- Default for files: `666` → `rw-rw-rw-`
- Default for directories: `777` → `rwxrwxrwx`

Formula: **Default - umask = actual permissions**

Check `umask`

```
$ umask
0022
```

♓Means: remove write from group & others.

Example

- File: `666 - 022 = 644` → `rw-r--r--`
- Dir: `777 - 022 = 755` → `rwxr-xr-x`

Set `umask`

```
$ umask 0077
```

♓New files → `600`, dirs → `700`. Only owner has access.

✅Summary

- `ls -l` shows permissions.

- `chmod` changes permissions.
- `umask` defines default permissions.
- Use numeric (`755`, `644`) or symbolic (`u+x`, `g-w`) modes.
- Be careful with `777` (full access to everyone).